

Calc 1 Lecture 22: Antiderivatives!

Motivating question: Suppose we know your velocity over time.

Can we use that to figure out your position at any given time?

Answer: **No.**

But if we also know your initial position,
then we can find your position at any time

Position $\xrightarrow{\text{deriv}}$ velocity
velocity $\xrightarrow{\text{antider.v.}}$ position

This motivates the following definition:

Definition: given a function $f(x)$, an antiderivative of $f(x)$
(over an interval I) is a function $F(x)$ satisfying
 $F'(x) = f(x)$ (over the interval I).

the derivative of F is $f \iff$ antiderivative of f is F

*note: antiderivatives are not unique.
many possible options.

Example: What are some antiderivatives of $f(x) = x$?

Want to find some function $F(x)$ such that $F'(x) = x$

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx}\left(\frac{1}{2}x^2\right) = \frac{1}{2} \cdot 2x = x \quad \checkmark \quad \Rightarrow F(x) = \frac{1}{2}x^2 \text{ is an antiderivative of } x$$

$$\text{e.g. } \frac{d}{dx}\left(\frac{1}{2}x^2 + 5\right) = \frac{d}{dx}\left(\frac{1}{2}x^2\right) = x \quad \checkmark \quad \Rightarrow F(x) = \frac{1}{2}x^2 + C$$

is an antiderivative of x
for any constant C

Observation:

- if $F(x)$ is an antiderivative of $f(x)$, then $F(x) + C$ is also an antiderivative of $f(x)$.
- if $F_1(x)$ and $F_2(x)$ are both antiderivatives of $f(x)$, then

$$F_2(x) = F_1(x) + C$$

$$\text{explanation: } \frac{d}{dx}(F_2 - F_1) = \underbrace{F_2'(x)}_{f(x)} - \underbrace{F_1'(x)}_{f(x)} = 0 \quad \text{for all } x$$

$$\Rightarrow F_2 - F_1 = C \Rightarrow F_2 = F_1 + C$$

In conclusion:

Theorem If F is an antiderivative of f on an interval I

then the most general antiderivative over I is of the form

$$F(x) + C$$

↑ the family of all possible antiderivatives

where C is an arbitrary constant.

Example: Find the general antiderivative of $f(x)$:

a) $f(x) = \sin(x)$

Step 1: Find a specific antiderivative $F(x)$

$$\frac{d}{dx}(F) = \sin x ?$$

$$\frac{d}{dx}(\cos x) = -\sin x \Rightarrow \frac{d}{dx}(-\cos x) = \sin x$$

$$\Rightarrow \boxed{F(x) = -\cos x}$$

Step 2: General antiderivative is

$$\boxed{-\cos(x) + C}$$

b) $f(x) = \frac{3}{\sqrt{x}}$

$$\frac{d}{dx}(F) = \frac{3}{\sqrt{x}} = 3 \cdot x^{-1/2}$$

$$\frac{d}{dx}(x^{1/2}) = \frac{1}{2} \cdot x^{-1/2}$$

↓ $\cdot 6$

$$\frac{d}{dx}(6x^{1/2}) = 3x^{-1/2} \Rightarrow$$

$$F(x) = 6x^{1/2} \quad \text{specific solution}$$

$$F(x) = 6x^{1/2} + C \quad \text{general solution}$$

c) $f(x) = 2^x$

$$\frac{d}{dx}(F) = 2^x \quad ?$$

$$\frac{d}{dx}(2^x) = \ln(2) \cdot 2^x$$

↓ divide by $\ln(2)$

$$\frac{d}{dx}\left(\frac{2^x}{\ln(2)}\right) = 2^x \quad \} \Rightarrow$$

$$\boxed{\frac{2^x}{\ln(2)} + C}$$

general solution

Observation: antiderivative of x^n ? (for $n \neq -1$)

$$\frac{d}{dx}(x^{n+1}) = (n+1)x^n$$

$\Rightarrow$ antiderivative of x^n

↓ divide by $n+1$

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$$

is

$$\boxed{\frac{x^{n+1}}{n+1} + C}$$

• if $n = -1$: $f(x) = x^{-1} = \frac{1}{x} \Rightarrow F(x) = \ln|x| + C$

Some more examples: Find the general antiderivative

a) $f(x) = x(\sqrt{x} + 3)$

first simplify!

$$f(x) = \underbrace{x \cdot \sqrt{x}}_{x^{3/2}} + \underbrace{3x}_{x^1} = \underbrace{x^{3/2}}_{\substack{\text{antider.} \\ \downarrow \\ x^{5/2}}} + 3 \underbrace{x^1}_{\substack{\downarrow \\ x^2/2}}$$

5/2

$$F(x) = \frac{x^{5/2}}{5/2} + 3 \frac{x^2}{2} + C = \boxed{\frac{2}{5} x^{5/2} + \frac{3}{2} x^2 + C}$$

check our work: $\frac{d}{dx} F \stackrel{?}{=} f$

$$\begin{aligned} \frac{d}{dx} \left(\frac{2}{5} x^{5/2} + \frac{3}{2} x^2 \right) &= \cancel{\frac{2}{5}} \cdot \cancel{\frac{5}{2}} x^{3/2} + \frac{3 \cdot 2x}{2} \\ &= x^{3/2} + 3x = f(x) \quad \checkmark \end{aligned}$$

e) $\sin(2x)$

want F such that $\frac{d}{dx}(F) = \sin(2x)$

$$\text{try: } \frac{d}{dx}(\cos(2x)) = -\sin(2x) \cdot 2$$

$\Downarrow$ divide by -2

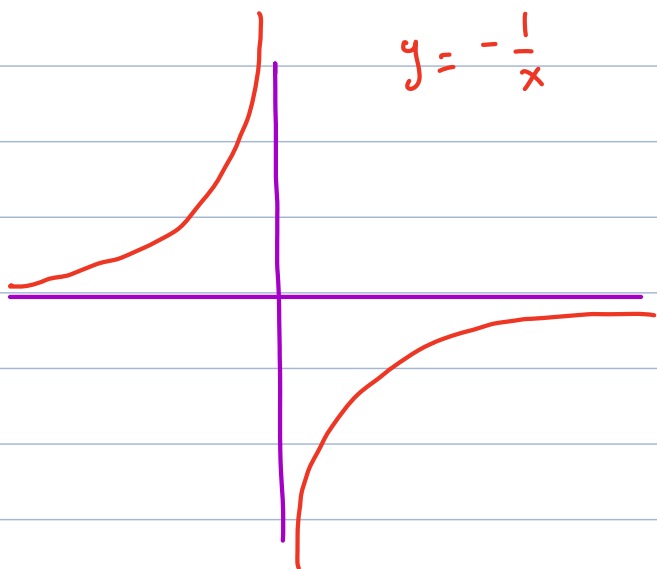
$$\frac{d}{dx} \left(\underbrace{\frac{-\cos(2x)}{2}}_F \right) = \sin(2x)$$

$$\boxed{F = \frac{-\cos(2x)}{2} + C}$$

Caution: the result that
 $(\text{general antideriv}) = (\text{specific antideriv.}) + C$
is only valid over an interval.

Example a) What is the general antiderivative of $f(x) = \frac{1}{x^2} = x^{-2}$
over the entire domain?

specific antideriv: $\frac{x^{-1}}{-1} = \left(-\frac{1}{x}\right)$



general antiderivative:

$$\begin{cases} -\frac{1}{x} + C_1 & \text{if } x < 0 \\ -\frac{1}{x} + C_2 & \text{if } x > 0 \end{cases}$$

b) What is the general antiderivative of $f(x) = \frac{1}{x}$?

specific: $\ln|x|$

general:

$$\begin{cases} \ln|x| + C_1 & \text{if } x < 0 \\ \ln|x| + C_2 & \text{if } x > 0 \end{cases}$$

First table of antiderivatives

derivative

Function

Antiderivative (specific)

c (constant)

cX

x^n ($n \neq -1$)

$\frac{x^{n+1}}{n+1}$

$\frac{1}{x}$

$\ln|x|$

e^x

e^x

b^x

$\frac{b^x}{\ln(b)}$

$\cos(x)$

$\sin(x)$

$\sin(x)$

$-\cos(x)$

$\sec^2(x)$

$\tan(x)$

$\sec x \tan(x)$

$\sec(x)$

$\frac{1}{\sqrt{1-x^2}}$

$\sin^{-1}(x)$

$\frac{1}{1+x^2}$

$\tan^{-1}(x)$

Some further anti derivative rules:

Suppose F is antiderivative of f ($F' = f$)
 G is antiderivative of g ($G' = g$)

- (Sum) $F + G$ is an antiderivative of $f + g$

$$\left(\frac{d}{dx} (F + G) = F' + G' = f + g \right)$$

- (constant multiple) cF is an antiderivative of cf

$$\left(\frac{d}{dx} (cF) = cF' = cf \right)$$

- don't have a product rule or chain rule

- antiderivative of $f(cx)$ is $\frac{F(cx)}{c}$

$$\left(\frac{d}{dx} \left(\frac{F(cx)}{c} \right) = \frac{\cancel{F'(cx)} \cdot \cancel{c}}{\cancel{c}} \right\} \text{chain} = F'(cx) = f(cx)$$

e.g. antiderivative e^{3x}

antiderivative of e^x is e^x

antiderivative of e^{3x} is $\frac{e^{3x}}{3}$

Initial Value Problems

In initial value problems, we are given the derivative of a function ($f'(x)$) and its value at some specific point ($f(a)$) and we need to recover the function ($f(x)$)

Example Find $f(x)$ if $f'(x) = e^x + \frac{20}{1+x^2}$ and $f(0) = -2$.

f' is derivative of $f \Leftrightarrow f$ is an antiderivative of f'

Step 1a. Find specific antiderivative of $f'(x)$

antiderivative of $e^x + \frac{20}{1+x^2} = e^x + 20 \cdot \frac{1}{1+x^2}$

is $F(x) = e^x + 20 \tan^{-1} x$

Step 1b. Find general antiderivative:

$$f(x) = e^x + 20 \tan^{-1} x + C \quad \text{for some constant } C$$

Step 2a. Use initial value to solve for

$$f(0) = -2 \quad \rightarrow \quad \underbrace{e^0}_1 + 20 \underbrace{\tan^{-1}(0)}_0 + C = -2$$

$$1 + 0 + C = -2 \Rightarrow \boxed{C = -3}$$

Step 2b. plug back in to get $f(x)$

$$f(x) = e^x + 20 \tan^{-1} x + C = \boxed{e^x + 20 \tan^{-1} x - 3 = f(x)}$$

We can also find $f(x)$ if we are given $f''(x)$ and two pieces of information (either about f or f')

Example Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, $f(1) = 1$

$$f'' \xRightarrow{\text{antideriv.}} f' \xRightarrow{\text{antideriv.}} f$$

Step 1: Find general form of f' (antiderivative of f'')

$$f''(x) = 12x^2 + 6x - 4 \Rightarrow \text{antideriv. is}$$

$$f'(x) = 12 \frac{x^3}{3} + 6 \frac{x^2}{2} - 4x + C = 4x^3 + 3x^2 - 4x + C$$

Step 2: Find general form of f (antiderivative of f')

$$f'(x) = 4x^3 + 3x^2 - 4x + C \Rightarrow \text{antideriv. is}$$

$$f(x) = \cancel{4} \frac{x^4}{\cancel{4}} + \cancel{3} \frac{x^3}{\cancel{3}} - 4 \frac{x^2}{2} + Cx + D = \boxed{x^4 + x^3 - 2x^2 + Cx + D} \quad f(x)$$

Step 3: Use initial values to solve for constants

$$f(0) = 4 \Rightarrow 0 + 0 + 0 + 0 + D = 4 \Rightarrow \boxed{D = 4}$$

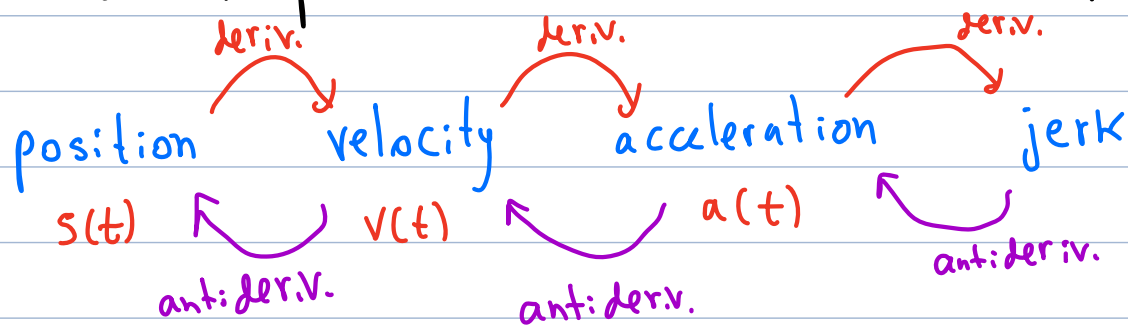
$$f(1) = 1 \Rightarrow \underbrace{1 + 1 - 2}_0 + C + D = 1 \Rightarrow C + D = 1 \Rightarrow \begin{aligned} C + 4 &= 1 \\ \Rightarrow \boxed{C = -3} \end{aligned}$$

Step 4: plug constants in to general form to get specific form

$$\boxed{f(x) = x^4 + x^3 - 2x^2 - 3x + 4}$$

Motion

Similar examples arise in contexts of motion.



Example You drop a ball from a height of 10 ft.
How long will it take the ball to reach the ground?

Gravity on earth: acceleration of $\sim -32 \frac{\text{ft}}{\text{s}^2}$

$$a(t) = -32$$

$$v(t) = -32t + C$$

$$s(t) = -32 \frac{t^2}{2} + Ct + D$$

anti-deriv.

anti-deriv.

(s represents vertical height)

what are C and D?

given information:

- $s(0) = 10$ (define time 0 as time when you drop ball)

$$0 + 0 + D = 10 \Rightarrow D = 10$$

- $v(0) = 0$

$$0 + C = 0 \Rightarrow C = 0$$

$$s(t) = -32 \frac{t^2}{2} + \cancel{Ct} + D = -\frac{32t^2}{2} + 10 = -16t^2 + 10$$

$s(t)$

Question: how long will it take the ball to reach
the ground?
 $s(t) = 0$

$$-16t^2 + 10 = 0 \quad ?$$

$$16t^2 = 10$$

$$t^2 = \frac{10}{16} = \frac{5}{8} \Rightarrow$$

$$t = \sqrt{\frac{5}{8}} \text{ seconds}$$

Example A car braked with constant deceleration 16 ft/sec^2
negative acceleration

producing skid marks measuring 200 ft before coming to a stop.

How fast was the car traveling when the brakes were first applied? $t=0$

given: $a(t) = -16$ $\left. \begin{array}{l} \text{ant: der. v.} \end{array} \right\}$

$v(t) = -16t + C$ $\left. \begin{array}{l} \text{ant: der. v.} \end{array} \right\}$

$s(t) = -16 \frac{t^2}{2} + Ct + D$

Convention: $t=0$ is when you first hit the brakes

i) $s(0) = 0$.

" skid marks measuring 200 ft :

let t_1 represent time when you come to a stop.

ii) $v(t_1) = 0$ (since you come to a stop)

iii) $s(t_1) = 200$

Want to find: $v(0) = C = ?$

i) $s(0) = 0 \Rightarrow 0 + 0 + D = 0 \Rightarrow D = 0$.

ii) $v(t_1) = 0 \Rightarrow -16t_1 + C = 0 \Rightarrow \underline{C = 16t_1}$

iii) $s(t_1) = 200 \Rightarrow -16 \frac{t_1^2}{2} + \underbrace{Ct_1}_{16t_1} + \underbrace{D}_0 = 200$

$$\frac{-16 t_1^2}{2} + 16 t_1 \cdot t_1 = 200$$

$$-8 t_1^2 + 16 t_1^2 = 200$$

$$8 t_1^2 = 200$$

$$t_1^2 = \frac{200}{8} = 25$$

$$\underline{t_1} = \sqrt{25} = \boxed{5} \text{ seconds}$$

$$\underline{C} = 16 t_1 = 16 \cdot 5 = 80$$

$$\Rightarrow \boxed{v(0) = C = 80 \frac{\text{ft}}{\text{sec}}}$$

